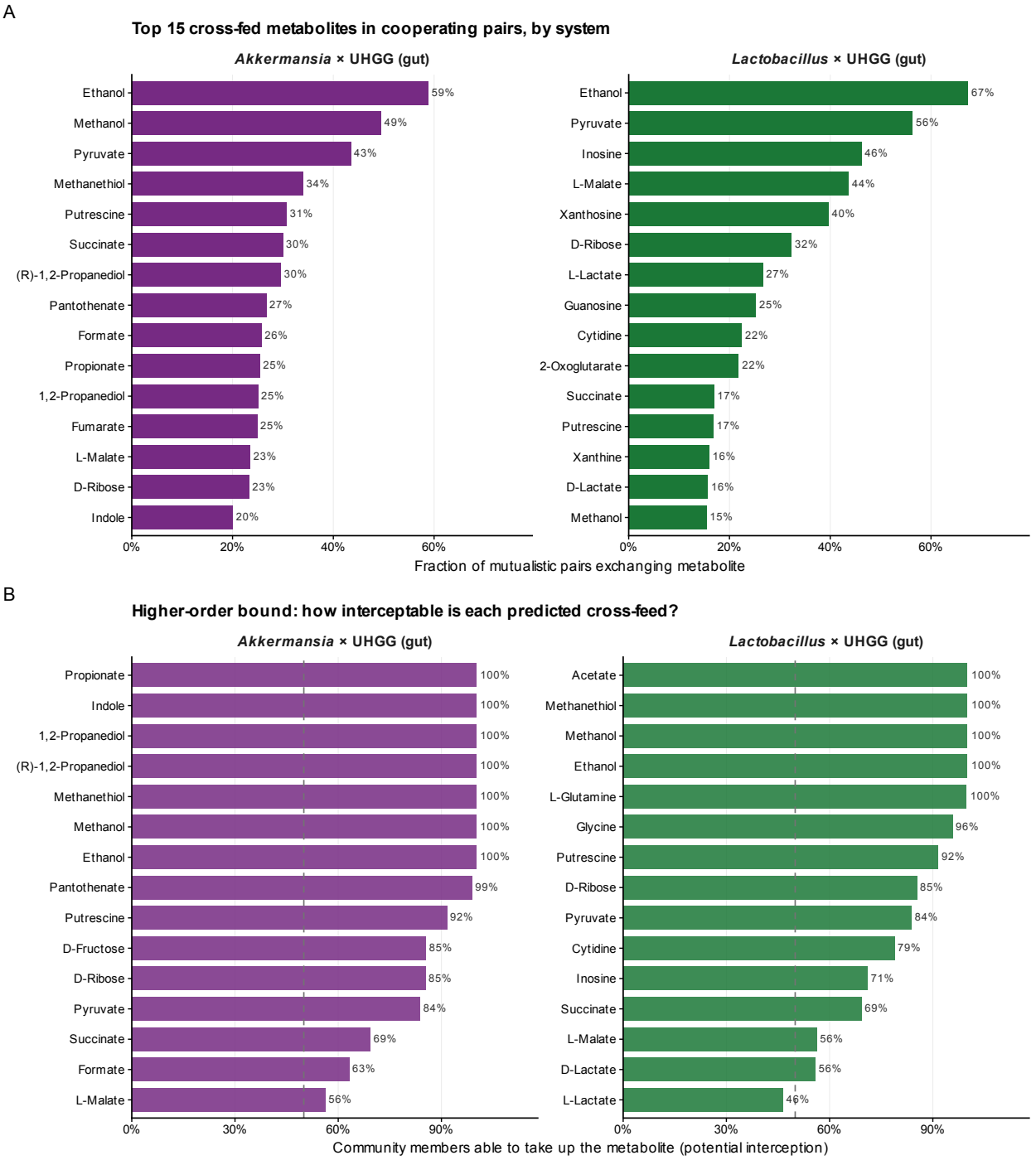
A
Top 15 cross-fed metabolites in cooperating pairs, by system
Akkermansia × UHGG (gut)
Ethanol 59%
Methanol 49%
Pyruvate 43%
Methanethiol 34%
Putrescine 31%
Succinate 30%
(R)-1,2-Propanediol 30%
Pantothenate 27%
Formate 26%
Propionate 25%
1,2-Propanediol 25%
Fumarate 25%
L-Malate 23%
D-Ribose 23%
Indole 20%
0% 20% 40% 60%
Lactobacillus × UHGG (gut)
Ethanol 67%
Pyruvate 56%
Inosine 46%
L-Malate 44%
Xanthosine 40%
D-Ribose 32%
L-Lactate 27%
Guanosine 25%
Cytidine 22%
2-Oxoglutarate 22%
Succinate 17%
Putrescine 17%
Xanthine 16%
D-Lactate 16%
Methanol 15%
0% 20% 40% 60%
Fraction of mutualistic pairs exchanging metabolite
B
Higher-order bound: how interceptable is each predicted cross-feed?
Akkermansia × UHGG (gut)
Propionate 100%
Indole 100%
1,2-Propanediol 100%
(R)-1,2-Propanediol 100%
Methanethiol 100%
Methanol 100%
Ethanol 100%
Pantothenate 99%
Putrescine 92%
D-Fructose 85%
D-Ribose 85%
Pyruvate 84%
Succinate 69%
Formate 63%
L-Malate 56%
0% 30% 60% 90%
Lactobacillus × UHGG (gut)
Acetate 100%
Methanethiol 100%
Methanol 100%
Ethanol 100%
L-Glutamine 100%
Glycine 96%
Putrescine 92%
D-Ribose 85%
Pyruvate 84%
Cytidine 79%
Inosine 71%
Succinate 69%
L-Malate 56%
D-Lactate 56%
L-Lactate 46%
0% 30% 60% 90%
Community members able to take up the metabolite (potential interception)